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Image forming apparatus including a housing and drive unit and a driver disposed outside the housing

Abstract

An image forming apparatus includes a housing, a driven unit, and a driver. The housing includes a pair of plates facing each other. The pair of plates includes a front plate and a rear plate facing the front plate and having an opening. The driven unit is supported by the pair of plates and has one end projecting outside the housing from the opening of the rear plate. The driver is disposed outside the housing and coupled to said one end of the driven unit to drive the driven unit. The driver includes a body separated from an outer face of the rear plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This patent application is based on and claims priority pursuant to 35 U.S.C. § 119(a) to Japanese Patent Application No. 2022-190318, filed on Nov. 29, 2022, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.
BACKGROUND
Technical Field
(2) Embodiments of the present disclosure relate to an image forming apparatus.
Related Art
(3) One type of electrophotographic image forming apparatus such as a laser printer or a digital copier includes a photoconductor and a process unit including the photoconductor and units

arranged around the photoconductor to perform image forming operations of an electrophotographic process, such as a developing unit and a cleaning unit. The process unit is detachably attached to a main body of the image forming apparatus.
(4) The image forming apparatus includes a driver including a motor and a driving gear in the main body. The process unit includes a driven gear. When the process unit is installed in the main body of the image forming apparatus, the driven gear and the driving gear are connected to each other, and the driving force of the motor can be transmitted to the process unit, whereby the process unit is driven.
SUMMARY
(5) This specification describes an improved image forming apparatus that includes a housing, a driven unit, and a driver. The housing includes a pair of plates facing each other. The pair of plates includes a front plate and a rear plate facing the front plate and having an opening. The driven unit is supported by the pair of plates and has one end projecting outside the housing from the opening of the rear plate. The driver is disposed outside the housing and coupled to said one end of the driven unit to drive the driven unit. The driver includes a body separated from an outer face of the rear plate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:
(2) FIG. **1**A is a schematic diagram illustrating a configuration of an image forming apparatus according to an embodiment of the present disclosure;
(3) FIG. **1**B is a schematic cross-sectional side view of the image forming apparatus according to a comparative example viewed in a direction indicated by an arrow **1**B in FIG. **1**A;
(4) FIG. **1**C is a schematic cross-sectional side view of the image forming apparatus of FIG. **1**B and component arrangement spaces above and below a driver;
(5) FIG. **1**D is a schematic cross-sectional side view of the image forming apparatus having component arrangement spaces enlarged by moving a rear plate from the position illustrated in FIG. **1**C toward the front side;
(6) FIG. **2**A is a perspective view of a housing of the image forming apparatus obliquely viewed from the front of the image forming apparatus;
(7) FIG. **2**B is a perspective view of the housing of the image forming apparatus obliquely viewed from the rear of the image forming apparatus;
(8) FIG. **3**A is a schematic cross-sectional side view of the housing in which a process unit is installed;
(9) FIG. **3**B is a perspective view of the housing to which the process unit is installed obliquely viewed from the rear of the image forming apparatus;
(10) FIG. **4**A is a schematic cross-sectional side view of the housing of FIG. **3**A to which a bracket is attached;
(11) FIG. **4**B is a perspective view of the housing of FIG. **3**B to which the bracket is attached obliquely viewed from the rear of the image forming apparatus;
(12) FIG. **5**A is a schematic cross-sectional side view of the housing of FIG. **4**A to which a driver is attached;
(13) FIG. **5**B is a perspective view of the housing of FIG. **4**B to which the driver is attached obliquely viewed from the rear of the image forming apparatus;
(14) FIG. **6** is a perspective view of the rear plate on which support shafts are fixed by caulking;

(15) FIG. **7** is a perspective view of a bracket;
(16) FIG. **8**A is a perspective view of the bracket and screws to be screwed to the support shafts on the rear plate;
(17) FIG. **8**B is a perspective view of the bracket screwed to the support shafts on the rear plate;
(18) FIG. **9**A is a perspective view of a driver;
(19) FIG. **9**B is the perspective view of the bracket;
(20) FIG. **10** is a perspective view of the bracket to which the driver is attached, as viewed from the back side of the bracket;
(21) FIG. **11**A is a perspective view of one end of a process unit;
(22) FIG. **11**B is the perspective view of the driver;
(23) FIG. **12**A is a schematic cross-sectional side view of the housing of FIG. **1**D to which sheet ejection rollers and sheet conveyance rollers are attached;
(24) FIG. **12**B is a perspective view of the housing of FIG. **12**A obliquely viewed from the rear of the image forming apparatus;
(25) FIG. **13** is the schematic cross-sectional side view of the housing of FIG. **12**A to illustrate a space S**1** in the housing;
(26) FIG. **14**A is a schematic cross-sectional side view of the housing in which a fan duct unit is disposed by using the space S**1**;
(27) FIG. **14**B is a perspective view of the housing of FIG. **14**A obliquely viewed from the rear of the image forming apparatus;
(28) FIG. **15** is a front view of the rear plate to which the driver, the sheet ejection roller driver, and the sheet conveyance roller driver are attached;
(29) FIG. **16** is a schematic cross-sectional side view of the housing including the rear plate of FIG. **15** to illustrate a gear of the driver and a gear of the sheet conveyance roller driver that are meshed with each other;
(30) FIG. **17** is a schematic cross-sectional side view of the housing in which the driver is attached to the rear plate without using the bracket to illustrate an increase in a space regarding the driver;
(31) FIG. **18** is a schematic cross-sectional side view of the housing in which the driver is attached to the rear plate without using the bracket to illustrate an increase in a space regarding the fan duct;
(32) FIG. **19** is a perspective view of a waste toner discharge pipe and the one end of the process unit to be coupled to the waste toner discharge pipe;
(33) FIG. **20**A is a perspective view of the waste toner discharge pipe attached to an inner face of the bracket; and
(34) FIG. **20**B is a perspective view of a toner conveyance driver attached to an outer face of the bracket of FIG. **20**A.
(35) The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

(36) In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.
(37) Referring now to the drawings, embodiments of the present disclosure are described below. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.
(38) With reference to FIG. **1**A, an image forming apparatus **1** according to an embodiment of the present disclosure is described below. FIG. **1**A is a schematic diagram illustrating a configuration

of the image forming apparatus **1**. The image forming apparatus **1** includes a process unit **2** including a photoconductor **3**. The process unit is one example of a driven unit. In the following description, the image forming apparatus **1** is described as a monochrome image forming apparatus, but the image forming apparatus according to the present disclosure may be a color image forming apparatus.

(39) The image forming apparatus **1** includes the process unit **2** to form an image with black toner. The process unit **2** includes a photoconductor **3**, a charger, a developing device, and a cleaner and is detachably attached to the main body of the image forming apparatus **1**. The photoconductor **3** has a drum-shape.

(40) The image forming apparatus **1** includes a writing device **4**. The writing device **4** irradiates the photoconductor **3** with laser light to form an electrostatic latent image on an image forming region of the surface of the photoconductor **3**. Additionally, the image forming apparatus **1** includes a toner bottle **5** containing a developing toner above the writing device **4**.

(41) The image forming apparatus **1** includes a transfer device **6** including a transfer roller **7**. The transfer roller **7** contacts the photoconductor **3** to form a transfer nip. A transfer bias is applied to the transfer roller **7** to form a transfer electric field in the transfer nip.

(42) The image forming apparatus **1** includes a fixing device **8** above the transfer device **6** in FIG. **1**A. The fixing device **8** fixes a toner image transferred to a recording medium such as a sheet onto the recording medium. The fixing device **8** includes a heat roller having a heat generator inside the heat roller. The heat roller serves as a fixing roller. The fixing device also includes a pressure roller facing the fixing roller. Between the transfer device **6** and the fixing device **8**, the recording medium on which the toner image is transferred is conveyed toward the fixing device **8** along a conveyance direction of the recording medium.

(43) The image forming apparatus **1** includes a sheet feeding device **9** in a lower part of the image forming apparatus **1** to feed the recording medium to the transfer device **6**. The image forming apparatus **1** further includes a sheet ejection device **10** disposed on the left of the fixing device **8** in FIG. **1**A. The sheet ejection device **10** ejects the recording medium, which has passed through the fixing device **8**, to the outside of the image forming apparatus **1**.

(44) The image forming apparatus **1** includes a housing. The structure of the housing is described below. FIG. **1**D is a schematic cross-sectional side view of the image forming apparatus viewed in a direction indicated by an arrow **1**B in FIG. **1**. As illustrated in FIG. **1**D, the housing includes a pair of plates (that is, a front plate **11** and a rear plate **12**). A bracket **14** as a positioner is attached to the rear plate **12** as described below. On an outer face of the bracket **14**, a driver **15** is positioned and fixed. The driver **15** includes a motor and a drive gear to transmit a driving force of the motor to the process unit **2**. Alternatively, a driver may include a body including the motor and the drive gear and multiple leg portions integrally formed with the body and directly fixed and positioned to the rear plate **12**. In this case, the body of the driver is the driver **15** illustrated in the drawings.

(45) FIGS. **1**B and **1**C are schematic cross-sectional side views of an image forming apparatus according to a comparative example viewed in the direction indicated by the arrow **1**B in FIG. **1**A. The driver **15** is positioned and fixed on an outer face of a rear plate **12**′ of a housing of the image forming apparatus according to the comparative example. The position of the rear plate **12** in the present embodiment is shifted from the position of the rear plate **12**′ of the comparative example illustrated in FIG. **1**C toward the inside of the housing by a length S**2**. The range having the length S**2** is outside the photoconductor **3** as an image bearer. The range is also a non-sheet passing region (in other words, a non-image forming region), which does not affect image formation. The range having the length S**2** may be outside the image forming region in which the image is formed on the photoconductor as the image bearer in an axial direction of the photoconductor.

(46) Moving the rear plate **12** inward by the length S**2** can reduce the size of the housing of the image forming apparatus in the front-back direction by the length S**2**. Moving the rear plate **12** inward causes a long portion of one end of the process unit **2** to project from the outer face of the

rear plate **12** (in other words, the right side of the rear plate **12** in FIG. **1**D) toward the outside. As a result, the driver **15** to drive the process unit **2** is separated from the rear plate **12**.
(47) The driver **15** according to the comparative example is directly positioned and fixed to the rear plate **12′** of the housing as illustrated in FIG. **1**C. In this case, a space from the end of the photoconductor indicated by a dashed line to the rear plate **12′** in FIG. **1**C is outside the image formation area but is not enough to arrange various mechanisms and structures. The mechanisms and structures of devices to form the image, such as the sheet feeding device, the transfer device, or the fixing device cannot be arranged in the space from the end of the photoconductor indicated by the dashed line to the rear plate **12′**. As a result, dead spaces **7***a* and **7***b* are formed.
(48) Although component arrangement spaces **8***a* and **8***b* are formed outside the dead spaces **7***a* and **7***b* (that is, outside the rear plate **12′**) as illustrated in FIG. **1**C, the rear plate **12′** moved from the position of the rear plate **12** in FIG. **1**D to the outside of the image forming apparatus reduces the size of the component arrangement spaces **8***b* and **8***a*. As a result, the components that can be arranged in the component arrangement spaces **8***a* and **8***b* are restricted, or the size of the image forming apparatus increases.
(49) In the present embodiment, moving the rear plate **12** inward as illustrated in FIG. **1**D results in the separation of the driver **15** from the rear plate **12** as described above. For this reason, the driver **15** is positioned and fixed to the rear plate **12** via the bracket **14**.
(50) The above-described structure in the present embodiment can form the component arrangement spaces **8***a′* and **8***b′* having a margin larger than the component arrangement spaces **8***a* and **8***b* in the comparative example above and below the driver **15**. These component arrangement spaces **8***a′* and **8***b′* have larger margins in the front-back direction that is the left-right direction of FIG. **1**D than the component arrangement spaces **8***a* and **8***b* in FIG. **1**C. As a result, various components related to image formation can be arranged in the component arrangement spaces **8***a′* and **8***b′*, which can reduce the size of the image forming apparatus **1**.
(51) The present embodiment is described below in detail with reference to FIGS. **2** to **20**. FIG. **2**A is a perspective view of the housing of the image forming apparatus **1** obliquely viewed from the front of the image forming apparatus **1**, and FIG. **2**B is a perspective view of the housing of the image forming apparatus **1** obliquely viewed from the rear of the image forming apparatus.
(52) The front plate **11** and the rear plate **12** are arranged to face each other. The front plate **11** has a rectangular opening **11***a*, and the rear plate **12** has a rectangular opening **12***a*. Between the openings **11***a* and **12***a*, a guide plate **13** is disposed to horizontally guide the process unit **2** inserted into the opening **11***a* of the front plate **11**.
(53) The front plate **11** has a support hole **11***c* above the opening **11***a* to support sheet ejection rollers and a support hole **11***b* below the opening **11***a* to support sheet conveyance rollers, and the rear plate **12** has a support hole **12***c* above the opening **12***a* to support the sheet ejection rollers and a support hole **12***b* below the opening **12***a* to support the sheet conveyance rollers.
(54) FIG. **3**A is a schematic cross-sectional side view of the housing to which the process unit **2** is installed, and FIG. **3**B is a perspective view of the housing to which the process unit **2** is installed obliquely viewed from the rear of the image forming apparatus. Moving the rear plate **12** inward, that is, toward the front plate **11** causes the long one end of the process unit **2** (in other words, a long rear end portion of the process unit **2**) to project from the outer face of the rear plate **12** (in other words, the right side of the rear plate **12** in FIG. **1**D) to the outside. The process unit **2** includes gears on the rear end portion of the process unit **2** to drive the photoconductor **3** and the developing device.
(55) The bracket **14** is described below. FIG. **4**A is a schematic cross-sectional side view of the housing of FIG. **3** in which the bracket **14** is attached to the rear plate **12**, and FIG. **4**B is a perspective view of the housing of FIG. **3** in which the bracket **14** is attached to the rear plate **12** obliquely viewed from the rear of the image forming apparatus. The bracket **14** is positioned and fixed on the rear plate **12** using two support shafts **16** illustrated in FIG. **6**.

(56) To position and support the driver **15**, the bracket **14** is positioned and attached to the rear plate **12** of the housing of the image forming apparatus with high accuracy. In the present embodiment, the two support shafts **16** can position the bracket **14** on the rear plate **12** with high accuracy.
(57) The bracket **14** has a size necessary and sufficient for mounting the driver **15** in the height direction and the width direction of the image forming apparatus **1**. Spaces on the outer face of the rear plate **12** that are not occupied by the bracket **14**, that is, the spaces above and below the bracket **14**, a space left from the bracket **14**, and a space right from the bracket **14** may be used to arrange other units. In addition, a space behind the driver **15** may be used to arrange the other units.
(58) The area of the bracket **14** is nearly twice as large as the projected area of the process unit **2** in the axial direction of the process unit **2** (in other words, a longitudinal direction of the process unit **2**) or the insertion and removal direction (that is indicated by an arrow A in FIG. **4**B) with respect to the bracket **14**. The area of the bracket **14** is also considerably larger than the area required to attach the driver **15** to the bracket **14**.
(59) The reason why the bracket **14** is made larger in this way is that a waste toner discharging pipe **24** and a drive gear **26***b* are positioned and fixed on the back side of the bracket **14** (in other words, an inner face of the bracket **14**) as described below with reference to FIGS. **19** and **20**. If the waste toner discharging pipe **24** and the drive gear **26***b* are not fixed on the bracket **14**, the size of the bracket **14** may be a minimum size to assemble the driver **15**.
(60) FIG. **5**A is a schematic cross-sectional side view of the housing to which the bracket **14** is attached, and the driver **15** is attached on the outer face of the bracket **14**. FIG. **5**B is a perspective view of the housing of FIG. **5**A obliquely viewed from the rear of the image forming apparatus. In FIG. **5**A, a right end of the driver **15** extends to the vicinity of a folded portion **12**A of the rear plate **12**. An exterior member is disposed outside the folded portion **12**A.
(61) The rear plate **12** in the present embodiment is arranged closer to the front plate **11** than the rear plate in the comparative example. As a result, a space upper the driver **15** in FIGS. **5**A and **5**B, a space below the driver **15** in FIGS. **5**A and **5**B, and a space from the right side of the driver **15** in FIG. **5**B each have sufficient room as the component arrangement space.
(62) The support shafts **16** are described below. FIG. **6** is a perspective view of the rear plate **12**. As illustrated in FIG. **6**, the two support shafts **16** are fixed on the rear plate **12** by caulking. The bases of the two support shafts **16** are extremely accurately caulked and fixed to the predetermined positions of the outer face of the rear plate **12**. The bracket **14** and the support shafts **16** serve as the positioner to position the driver **15**.
(63) The support shafts **16** are made of a steel material such as steel use stainless (SUS). The support shaft **16** is positioned and fixed to the rear plate **12** by using a caulking method, which enables positioning the support shaft **16** perpendicular to the rear plate **12** with high accuracy.
(64) As illustrated in an enlarged view of FIG. **6**, the support shaft **16** has a tip cylindrical portion **16***a* and a female screw hole **16***b* for screw fastening on a distal end of the support shaft **16**. The tip cylindrical portions **16***a* of the two support shafts are fitted to support shaft fitting holes **14***a* and **14***b* of the bracket **14**.
(65) The height of the tip cylindrical portion **16***a* is substantially equal to the thickness of a plate-shaped body of the bracket **14**. The female screw hole **16***b* is formed in the axis of the tip cylindrical portion **16***a*. Fastening screws **17** into the female screw holes **16***b* as illustrated in FIGS. **8**A and **8**B fixes the bracket **14** to the support shafts **16** and thus fixes the bracket **14** at a predetermined position on the rear plate **12**.
(66) The two support shafts **16** are obliquely arranged across the rectangular opening **12***a* of the rear plate **12**. In other words, the two support shafts **16** are arranged across a diagonal line of the rectangular opening **12***a*. In addition, the two support shafts **16** are on a diagonal line of the plate-shaped body of the bracket **14**. One end of the process unit **2** passes through the opening **12***a* and projects from the rear plate **12** toward the outside of the image forming apparatus (in other words,

to the rearward of the image forming apparatus).
(67) The two support shafts **16** are obliquely arranged in order to make the distance between the support shafts **16** as long as possible. Using the two support shafts **16** which are separated from each other by a long distance to position and fix the bracket **14** increases the positioning accuracy of the bracket **14** and the positioning accuracy of the driver **15**. The bracket is described below.
(68) FIG. **7** is a perspective view of the bracket **14**. The bracket **14** includes the plate-shaped body having a horizontally long rectangular shape and L-shaped leg portions **14***g* to **14***k* integrally formed on the sides of the plate-shaped body.
(69) Support shaft fitting holes **14***a* and **14***b* are formed at two corners of the plate-shaped body on a diagonal line of the plate-shaped body. Each of the tip cylindrical portions **16***a* of the support shafts **16** are fitted into any one of the support shaft fitting holes **14***a* and **14***b*.
(70) The support shaft fitting hole **14***a* is a round hole, but the support shaft fitting hole **14***b* is an elongated hole. The elongated hole of the support shaft fitting hole **14***b* extends toward the support shaft fitting hole **14***a* in a diagonal direction.
(71) The support shaft fitting hole **14***a* that is the round hole serves as a first positioning hole as a main reference for positioning, and the support shaft fitting hole **14***b* that is the elongated hole serves as a second positioning hole as a sub-reference for positioning. Fitting the tip cylindrical portion **16***a* of the support shaft **16** to the support shaft fitting hole **14***b* that is the elongated hole determines a position of the bracket **14** in a rotation direction around the support shaft fitting hole **14***a*.
(72) The height of the tip cylindrical portion **16***a* is substantially equal to the thickness of the plate-shaped body of the bracket **14**. The female screw hole **16***b* is formed in the axis of the tip cylindrical portion **16***a*. Fastening screws **17** into the female screw holes **16***b* as illustrated in FIGS. **8**A and **8**B fixes the bracket **14** to the support shafts **16** and thus fixes the bracket **14** at a predetermined position on the rear plate **12**.
(73) As illustrated in FIG. **9**B, the plate-shaped body of the bracket **14** has holes **14***c* to **14***f*. As illustrated in FIG. **9**A, the driver **15** has a circular projection **15***a*, a small cylinder **15***b*, an elliptical projection **15***d*, and a cylindrical developing roller driving gear **15***f* each of which passes through any one of the holes **14***c* to **14***f*. The corresponding relationship of the above-described parts and holes is illustrated in FIGS. **9**A and **9**B.
(74) The elliptical projection **15***d* has an elongated hole extending toward a photoconductor drive gear **15***e*. As illustrated in FIGS. **11**A and **11**B, a projection **2***d* of the process unit **2** is fitted into the elongated hole inside the elliptical projection **15***d*.
(75) The photoconductor drive gear **15***e* meshes with a photoconductor drive gear **2***e* of the process unit **2**. As a result, in the positioning relationship between the process unit **2** and the driver **15**, the photoconductor drive gear **15***e* serves as a main reference for positioning, and the elliptical projection **15***d* serves as a sub-reference for positioning.
(76) It is desirable that the circular projection **15***a* as a first projection of the driver **15** is fitted to the hole **14***c* of the bracket **14** with a minimum dimensional tolerance. Fitting the circular projection **15***a* of the driver **15** to the hole **14***c* of the bracket **14** with the minimum dimensional tolerance accurately aligns the axis of the photoconductor drive gear **15***e* of the driver **15** with the hole **14***c* of the bracket **14**.
(77) The small cylinder **15***b* as a second projection of the driver **15** is fitted into a hole **14***f* of the bracket **14**. The hole **14***f* is an elongated hole extending toward the hole **14***c*.
(78) The hole **14***c* is a round hole serves as a third positioning hole as a main reference for positioning, and the hole **14***f* that is the elongated hole serves as a fourth positioning hole as a sub-reference for positioning. Fitting the small cylinder **15***b* of the driver **15** into the hole **14***f* of the bracket **14** determines a position of the driver **15** in a rotation direction around the hole **14***c* that is the round hole and serves as the main reference of the bracket **14** for positioning.
(79) The photoconductor drive gear **15***e* is concentrically disposed with the circular projection **15***a*

of the driver **15** and projects from the circular projection **15***a*. The photoconductor drive gear **15***e* meshes with the photoconductor drive gear **2***e* of the process unit **2** in FIG. **11** to transmit the driving force from the driver **15** to the process unit **2**.

(80) Between the circular projection **15***a* and the photoconductor drive gear **15***e*, a circular projection **15***c* is disposed. The circular projection **15***c* is fitted into the inside of a cylinder **2***c* formed on the outer periphery of the photoconductor drive gear **2***e* of the process unit **2** illustrated in FIG. **11**A. An interdigitation of the circular projection **15***c* and the cylinder **2***c* can enhance the concentricity of the photoconductor drive gears **15***e* and **2***e*.

(81) As illustrated in FIG. **9**A, the driver **15** has screw holes **15***m* at four positions on the left, right, top and bottom of the driver **15**. As illustrated in FIG. **9**B, the bracket **14** has screw holes **14***m* at four positions of the plate-shaped body of the bracket **14**.

(82) As illustrated in FIG. **10**, four screws **14***n* are screwed into the four screw holes **14***m* of the bracket **14** from the back side of the bracket **14**, pass through the plate-shaped body of the bracket **14**, and are screwed into the screw holes **15***m* of the driver **15**.

(83) The driver **15** is positioned and fixed to the bracket **14** as illustrated in FIG. **10**, and the bracket **14** is positioned and fixed to the rear plate **12**. Subsequently, the process unit **2** is inserted into the opening **11***a* of the front plate **11** and slides along the guide plate **13**. Then, the photoconductor drive gear **2***e* meshes with the photoconductor drive gear **15***e*, which can be seen from FIGS. **11**A and **11**B. Parts of the process unit **2** fit to parts of the driver **15** as follows.

(84) The photoconductor drive gear **2***e* meshes with the photoconductor drive gear **15***e*.

(85) The circular projection **15***c* is fitted into the inside of the cylinder **2***c*.

(86) The projection **2***d* is fitted into the elongated hole inside the elliptical projection **15***d*.

(87) A developing roller driving gear **2***f* meshes with the developing roller driving gear **15***f*.

(88) FIG. **12**A is a schematic cross-sectional side view of the housing of FIG. **1**D to which sheet ejection rollers **18** and sheet conveyance rollers **19** are attached, and FIG. **12**B is a perspective view of the housing of FIG. **12**A obliquely viewed from the rear of the image forming apparatus. As illustrated in FIGS. **12**A and **12**B, the spaces around the driver **15** are effectively used as the component arrangement spaces. A sheet ejection roller driver **20** to drive the sheet ejection rollers **18** is in the space above the driver **15**. A sheet conveyance roller driver **21** to drive the sheet conveyance rollers **19** is in the space below the driver **15**. However, these are examples, and the usage of spaces around the driver **15** is not limited to these.

(89) As illustrated in FIGS. **12**A and **12**B, the sheet ejection roller driver **20** to drive the sheet ejection rollers **18** may be disposed above the bracket **14** and attached to the rear plate **12**, and the sheet conveyance roller driver **21** to drive the sheet conveyance rollers **19** may be disposed below the bracket **14** and attached to the rear plate **12**. A unit such as the sheet ejection roller driver **20** or the sheet conveyance roller driver **21** to drive a mechanism disposed between the front plate **11** and the rear plate **12**, such as the sheet ejection rollers **18** or the sheet conveyance rollers **19** may be directly attached to the outer face of the rear plate **12**.

(90) A space S**1** produced by the present disclosure is described below. FIG. **13** is the schematic cross-sectional side view of the housing of FIG. **12**A to illustrate the space S**1** formed behind the bracket **14**. A component may be disposed in the space S**1** behind the bracket **14** (in other words, the space S**1** from a right side of the bracket **14** in FIG. **13**).

(91) Another unit may be disposed in the space S**1** between an end of the driver **15** and the sheet ejection roller driver **20** in the axial direction of the photoconductor as illustrated in FIG. **13**. For example, as illustrated in FIGS. **14**A and **14**B, a fan duct unit **23** to exhaust heat from the fixing device **8** and the sheet ejection device **10** may be disposed in the space S**1**.

(92) The above-described fan duct unit **23** has a relatively large thickness in the front-back direction of the housing. It is not necessarily easy to dispose the fan duct unit **23** in the component arrangement space **8***a* or **8***b* formed by the position of the rear plate **12′** according to the comparative example illustrated in FIG. **1**C. In contrast, moving the rear plate **12** toward the front

side of the housing can increase the size of the component arrangement spaces **8***a*′ and **8***b*′ in the front-back direction as illustrated in FIG. **1**D. As a result, even a component having the relatively large thickness in the front-back direction, such as the fan duct unit **23**, can be easily disposed in the present embodiment.

(93) The following describes a relationship between a sheet conveyance roller drive gear and the size of the image forming apparatus. FIG. **15** is a front view of the rear plate **12** to which the driver **15**, the sheet ejection roller driver **20**, and the sheet conveyance roller driver **21** are attached. FIG. **16** is a schematic cross-sectional side view of the housing including the rear plate of FIG. **15** to illustrate a gear of the driver and a gear of the sheet conveyance roller driver that are meshed with each other. To transmit the driving force from the driver **15** to the sheet conveyance roller driver **21**, the driver **15** includes a sheet conveyance roller drive gear **15***g*, and the sheet conveyance roller driver **21** includes a sheet conveyance roller drive gear **21***a* that meshes with the sheet conveyance roller drive gear **15***g* as illustrated in FIG. **15**. In the above-described structure, the position of the sheet conveyance roller drive gear **15***g* of the driver **15** is the same as the position of the sheet conveyance roller drive gear **21***a* of the sheet conveyance roller driver **21** in the axial direction of the photoconductor **3** as illustrated in FIG. **16**.

(94) In the image forming apparatus not including the bracket **14**, the driver **15**′ is attached to the rear plate **12**′ as illustrated in FIG. **17**. In the axial direction of the photoconductor **3**, the above-described structure shifts the position of the rear plate **12**′ from the position of the rear plate **12** in the image forming apparatus including the bracket **14** (that is indicated by a dashed line in FIG. **17**) to the right side in FIG. **17** by the dimension of the bracket **14** in the axial direction. In other words, a space S**2** is formed between the rear plate **12**′ and the rear plate **12** (that is indicated by a dashed line in FIG. **17**) in the present embodiment as illustrated in FIG. **17**.

(95) In the above-described structure, the sheet conveyance roller driver **21** attached to the rear plate **12**′ is also shifted rearward in the axial direction of the photoconductor by an amount corresponding to the space S**2** because the drive gears **15***g* and **21***a* are disposed at the same position in the axial direction. As a result, the size of the driver **15**′ in the axial direction of the photoconductor is increased by an amount of S**3** corresponding to the space S**2**, which leads to an increase in the size of the image forming apparatus **1** in the axial direction of the photoconductor.

(96) Similarly, in the structure including the driver **15**′ attached to the rear plate **12**′ without using the bracket **14** as illustrated in FIG. **18**, the sheet ejection roller driver **20** is attached to the rear plate **12**′, and the fan duct unit **23** is disposed behind the sheet ejection roller driver **20**. This means that the size of the image forming apparatus **1** in the axial direction of the photoconductor is increased by an amount of a space S**4** corresponding to the space S**2**.

(97) As described above, the structure including the driver **15**′ attached to the rear plate **12**′ without using the bracket **14** increases the size of the image forming apparatus **1**. In contrast, using the bracket **14** having a size necessary and sufficient for mounting the driver **15** in the height direction and the width direction of the image forming apparatus **1** and using the space behind the rear plate **12** which is not occupied by the bracket **14** for the arrangement of other units can contribute to the miniaturization of the image forming apparatus **1**.

(98) The following describes an arrangement of a pipe to correct waste toner. In an image forming process, toner remains on the photoconductor after the toner image is transferred from the photoconductor **3** to the recording medium such as the sheet. Such toner is referred to as waste toner. The waste toner is corrected and stored in a storage such as a bottle **25**. As illustrated in FIG. **19**, a pipe **24** is connected to the rear end of the process unit **2** to convey the waste toner from the process unit **2** to the bottle **25** and store the waste toner in the bottle **25**.

(99) The pipe **24** may be disposed between the rear plate **12** and the bracket **14** and attached to the inner face of the bracket **14** by screws as illustrated in FIGS. **20**A and **20**B. Drive gears **26***a* and **26***b* may transmit a driving force to a conveyor such as a screw conveyor in the pipe **24** to convey the waste toner in the pipe **24**.

(100) As illustrated in FIG. **20**B, a toner conveyance driver **27** as a driver to drive the drive gears **26***a* and **26***b* may be attached to the front side of the bracket **14**. Designing an area of the face of the bracket **14** to which the driver **15** is attached to be larger than a projected area of the process unit **2** in the lateral direction of the image forming apparatus **1** enables using a part of the area to attach the pipe **24** and the toner conveyance driver **27**. In particular, disposing the pipe **24** in the space between the rear plate **12** and the bracket **14** effectively utilizes the space, avoids the occurrence of a dead space, and enhances a reduction in the size of the image forming apparatus **1**.
(101) Although the embodiments of the present disclosure are described above, the present disclosure is not limited to the embodiments and can be applied to other embodiments by modification in various forms. For example, the driver **15** in the present embodiment is positioned and fixed to the rear plate **12** via the bracket **14**, but the driver **15** may be positioned and fixed without using the bracket **14**. The body of the driver **15** illustrated in FIG. **9**A may have multiple leg portions integrally formed with the body of the driver **15** to extend to the rear plate **12**, and the tips of the leg portions may be directly positioned and fixed to the rear plate **12**. In the present embodiment, the process unit **2** is described as the one example of the driven units, but the driven unit may be another unit such as the fixing device **8** including the fixing roller and the pressure roller, the transfer device **6**, or a cleaning unit that removes and collects toner remaining on the photoconductor.
(102) The following describes preferred aspects of the present disclosure.
(103) First Aspect
(104) In a first aspect, an image forming apparatus includes a housing, a driven unit, and a driver. The housing includes a pair of plates facing each other. The pair of plates includes a front plate and a rear plate facing the front plate and having an opening. The driven unit is supported by the pair of plates and has one end projecting outside the housing from the opening of the rear plate. The driver is disposed outside the housing and coupled to said one end of the driven unit to drive the driven unit. The driver includes a body separated from an outer face of the rear plate.
(105) Second Aspect
(106) In a second aspect, the driven unit in the image forming apparatus according to the first aspect includes an image bearer having an image forming region on which an image is formed, and said one end projecting from the opening is outside the image forming region in an axial direction of the image bearer.
(107) Third Aspect
(108) In a third aspect, the driven unit in the image forming apparatus according to the first aspect includes a fixing device including a fixing roller and a pressure roller facing the fixing roller.
(109) Fourth Aspect
(110) In a fourth aspect, the driven unit in the image forming apparatus according to the first aspect includes a transfer device.
(111) Fifth Aspect
(112) In a fifth aspect, the image forming apparatus according to any one of the first to fourth aspects further includes a positioner fixed to the outer face of the rear plate, and the body of the driver is fixed to the positioner.
(113) Sixth Aspect
(114) In a sixth aspect, the positioner in the image forming apparatus according to the fifth aspect includes a bracket and multiple shafts fixing the bracket to the outer face of the rear plate, and each of the multiple shafts has one end fixed to the outer face of the rear plate by caulking.
(115) Seventh Aspect
(116) In a seventh aspect, the bracket in the image forming apparatus according to the sixth aspect has a first positioning hole and a second positioning hole. The first positioning hole serves as a main reference and is coupled to another end of one of the multiple shafts. The second positioning hole serves as a sub-reference and is coupled to another end of another of multiple shafts. The

second positioning hole is an elongated hole extending toward the first positioning hole.
(117) Eighth Aspect
(118) In an eighth aspect, the bracket in the image forming apparatus according to the sixth aspect or the seventh aspect has a rectangular shape, and the first positioning hole and the second positioning hole are on a diagonal line of the rectangular shape.
(119) Ninth Aspect
(120) In a ninth aspect, the bracket in the image forming apparatus according to the eighth aspect has a third positioning hole and a fourth positioning hole. The third positioning hole serves as another main reference to position the driver. The fourth positioning hole serves as another sub-reference to position the driver. The fourth positioning hole is an elongated hole extending toward the third positioning hole.
(121) Tenth Aspect
(122) In a tenth aspect, the driver in the image forming apparatus according to the ninth aspect includes a first drive shaft, a first projection, and a second projection. The first drive shaft is coupled to the image bearer. The first projection is concentrical with the first drive shaft and fitted into the third positioning hole. The second projection is fitted into the fourth positioning hole.
(123) Eleventh Aspect
(124) In an eleventh aspect, the image forming apparatus according to any one of the first to tenth aspects further includes another driver to drive a conveyance roller to convey a sheet, and said another driver is disposed below the driver.
(125) Twelfth Aspect
(126) In a twelfth aspect, the image forming apparatus according to any one of the first to eleventh aspects further includes a fan duct disposed above the driver.
(127) Thirteenth Aspect
(128) In a thirteenth aspect, the image forming apparatus according to any one of the sixth to twelfth aspects further includes a pipe coupled to the driven unit. The driver is fixed to an outer face of the bracket, and the pipe is on an inner face of the bracket.
(129) The above-described embodiments are illustrative and do not limit the present invention. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present invention. Any one of the above-described operations may be performed in various other ways, for example, in an order different from the one described above.

Claims

1. An image forming apparatus, comprising: a housing including a pair of plates facing each other, the pair of plates including: a front plate; and a rear plate facing the front plate and having an opening; a driven unit including a rotator and a portion covering the rotator, wherein one end of the driven unit projects outside the housing from the opening of the rear plate; and a driver disposed outside the housing and coupled to the one end of the driven unit, the driver configured to drive the rotator and the driver including a body separated from an outer face of the rear plate.
2. The image forming apparatus according to claim 1, wherein the rotator is a photoconductor or a developing roller.
3. The image forming apparatus according to claim 1, wherein the driven unit further includes a fixing device including: a fixing roller; and a pressure roller facing the fixing roller.
4. The image forming apparatus according to claim 1, wherein the driven unit further includes a transfer device.
5. An image forming apparatus, comprising: a housing including a pair of plates facing each other, the pair of plates including: a front plate; and a rear plate facing the front plate and having an

opening; a driven unit, supported by the pair of plates, having one end projecting outside the housing from the opening of the rear plate; a driver disposed outside the housing and coupled to the one end of the driven unit to drive the driven unit, the driver including a body separated from an outer face of the rear plate; and a positioner fixed to the outer face of the rear plate, wherein the body of the driver is fixed to the positioner, the positioner includes: a bracket; and multiple shafts fixing the bracket to the outer face of the rear plate, and each of the multiple shafts has one end fixed to the outer face of the rear plate by caulking.

6. The image forming apparatus according to claim 5, wherein the bracket has: a first positioning hole, coupled to another end of one of the multiple shafts, as a main reference; and a second positioning hole, coupled to another end of another of the multiple shafts, as a sub-reference, and the second positioning hole is an elongated hole extending toward the first positioning hole.

7. The image forming apparatus according to claim 6, wherein the bracket has a rectangular shape, and the first positioning hole and the second positioning hole are on a diagonal line of the rectangular shape.

8. The image forming apparatus according to claim 7, wherein the bracket further has: a third positioning hole as another main reference to position the driver; and a fourth positioning hole as another sub-reference to position the driver, and the fourth positioning hole is an elongated hole extending toward the third positioning hole.

9. The image forming apparatus according to claim 8, wherein the driver includes: a first drive shaft coupled to the rotator; a first projection concentrical with the first drive shaft and fitted into the third positioning hole; and a second projection fitted into the fourth positioning hole.

10. The image forming apparatus according to claim 1, further comprising: another driver to drive a conveyance roller to convey a sheet, the another driver disposed below the driver.

11. The image forming apparatus according to claim 1, further comprising: a fan duct disposed above the driver.

12. The image forming apparatus according to claim 5, further comprising: a pipe coupled to the driven unit, wherein the driver is fixed to an outer face of the bracket, and the pipe is on an inner face of the bracket.

13. The image forming apparatus according to claim 5, wherein the rotator is a photoconductor or a developing roller.
